BioHackrXiv

BioHackathon series:
SWAT4HCLS Biohackathon 2026
Amsterdam, The Netherlands
Project 5

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SWAT4HCLS Biohackathon 2026: Template for the very long title

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Introduction

As part of the SWAT4HCLS Hackathon 2026, we report on Project 5. This project explores the use of Large Language Model (LLM)-based agents for automated reconciliation of domain-specific terminology with structured ontology concepts. Semantic interoperability remains a major bottleneck in achieving FAIR data integration across biomedical and life sciences domains. Mapping heterogeneous terminology to standardized identifiers from resources such as Wikidata and BioPortal is essential but typically relies on manual curation.This is a time-consuming and difficult to scale process. Recent advances in LLM-based agent systems provide new opportunities to automate this process by combining semantic similarity, contextual reasoning, and tool-based API interaction. In this project, we evaluate a multi-agent system designed to perform ontology mapping and apply it to real-world biomedical data derived from the Synodos NF2 project.

Methods

System architecture

We implemented a multi-agent pipeline for automated mapping of domain-specific terms to ontology concepts. The system integrates:

- Ontology recommendation and search
- Definition retrieval and enrichment
- LLM-based reasoning for semantic alignment
- Assignment of structured identifiers and SKOS relations

The architecture is implemented using LangChain, LangGraph, and a Deep Agent framework.

Multi-agent design

The system consists of three coordinated agents: - Ontlogoy Agent - BioPortal Agent - Orchestrating Agent

Ontology Agent The ontology agent selects relevant ontologies using the BioPortal recommender services.

BioPortal Agent The Bioportal agent is resposible for retrieving candidate concepts. This includes labels, synonyms, and definitions.

Orchestrating Agent The Orchestrating agent integrates results, evaluates candidates, and assigns: - Best matching identifier (Wikidata QID or ontology URI) - SKOS semantic relation (e.g., exactMatch, closeMatch) - Explanation for the mapping

Data source: Synodos NF2 dataset

We evaluated the system using a subset of data from the Synodos NF2 Drug Screening Dataset, hosted on Synapse.

Specifically, we used the processed dataset:

- DOI: https://doi.org/10.7303/syn6138237.1
- File: Synodos_DrugScreen_processed_data.tsv

Additional related raw datasets (single-agent and combination screening) were also provided and used as supporting sources of terminology.

Meeting information

```yaml
biohackathon_name: "SWAT4HCLS Biohackathon 2026"
biohackathon_url: "https://www.swat4ls.org/swat4hcls-biohackathon-2026/"
biohackathon_location: "Amsterdam, The Netherlands"
group: Project 5
git_url: "https://github.com/AJKellmann/SWAT4HCLS26_hack-5-llm-agents-term-reconci
```

Formatting

This document use Markdown and you can look at this tutorial.

Subsection level 2

Please keep sections to a maximum of only two levels.

Tables

Tables can be added in the following way, though alternatives are possible:

```
Table: Note that table caption is automatically numbered and should be
given before the table itself.

| Header 1 | Header 2 |
| -------- | -------- |
| item 1 | item 2 |
| item 3 | item 4 |
```

This gives:

Table 1: Note that table caption is automatically numbered and should be given before the table itself.

Header 1	Header 2
item 1	item 2
item 3	item 4

Figures

A figure is added with:

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![Caption for BioHackrXiv logo figure](./biohackrxiv.png)
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Figure 1: Caption for BioHackrXiv logo figure

Figures can be scaled by adding the width or height to the Markdown like this:

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Other main section on your manuscript level 1

Lists can be added with:

1. Item 1
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Citation Typing Ontology annotation

You can use CiTO annotations, as explained in this BioHackathon Europe 2021 write up and this CiTO Pilot. Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethodIn:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- citesAsDataSource: when you point the reader to a source of data which may explain a claim
- usesDataFrom: when you reuse somehow (and elaborate on) the data in the cited entity
- usesMethodIn
- citesAsAuthority
- citesAsEvidence
- citesAsPotentialSolution
- citesAsRecommendedReading
- citesAsRelated
- citesAsSourceDocument
- citesForInformation
- confirms
- documents
- providesDataFor
- obtainsSupportFrom
- discusses
- extends

- agreesWith
- disagreesWith
- updates
- citation: generic citation

Results

Discussion

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Acknowledgements

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ReferencesRecent advances in LLM-based agent systems provide new opportunities to automate this process by combining semantic similarity, contextual reasoning, and tool-based API interaction. In this project, we evaluate a multi-agent system designed to perform ontology mapping and apply it to real-world biomedical data derived from the Synodos NF2 project.

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- Assignment of structured identifiers and SKOS relations

The architecture is implemented using LangChain, LangGraph, and a Deep Agent framework.

Multi-agent design

The system consists of three coordinated agents: - agent 1 - System 2

-**Ontology Agent** Selects relevant ontologies using BioPortal recommender services

-**BioPortal Agent** Retrieves candidate concepts, including labels, synonyms, and definitions

-**Orchestrating Agent** Integrates results, evaluates candidates, and assigns: Best matching identifier (Wikidata QID or ontology URI) SKOS semantic relation (e.g., exactMatch, closeMatch) Explanation for the mapping

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biohackathon_location: "Amsterdam, The Netherlands"
group: Project 5
git_url: https://github.com/yourOrganization/your_report_repo
```

The `git_url:` field must have the link to the GitHub repository with your preprint (draft).

Author information

Information about the authors is given in the YAML format at the top of this template. For authors you provide their names, their affiliations. That is the minimum, but as BioHackrXiv is moving to a situation where more metadata is shared, and used by, for example, EuropePMC, adding additional information ie encouraged.

BioHackathons is about hacking together, and the minimal number of authors for reports is two. This makes a minimal example look like this:

```yaml
authors:
  - name: First Author
    affiliation: 1
  - name: Last Author
    affiliation: 2
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    index: 2
```

Author identifiers

Ideally, authors provide their ORCID identifier. For affiliations, It is added with the `orcid:` field. So, and author record would look like this:

```yaml
authors:
  - name: First Author
    affiliation: 1
    orcid: 0000-0000-0000-0000
```

Research Organization Registry identifiers

Matching the author identifier, the affiliations can be further specified with the Research Organization Registry (ROR) identifier. For example, this is the affiliation identifier can be added with the `ror:` field:

```yaml
affiliations:
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
```

Contributor Role Taxonomy

A last feature since is minimal support for the Contributor Role Taxonomy (CRediT). You can specify the role of authors in writing the report with the `role:` field. However, the authors are responsible for selection the right terms from CRediT. An example looks like this:

```yaml
authors:
  - name: First Author
    affiliation: 1
    orcid: 0000-0000-0000-0000
    role: Conceptualization, Writing - review & editing
```

A full examples

A full example then has this structure:

```
authors:
  - name: First Author
    affiliation: 1
    role: Writing - original draft
  - name: Last Author
    orcid: 0000-0000-0000-0000
    affiliation: 2
    role: Conceptualization, Writing - review & editing
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
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References

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, *32*(14), 2236–2238. https://doi.org/10.1093/bioinformatics/btw214 **[cito:citesAsAuthority]**

Author contributions

Iurii Savvateev: Software, validation Woodward Galbraith: Software, validation Linda Hendriks: Writing – original draft, validation Alexander Kellmann: Writing – original draft, validation Nalini Paijens: Writing – original draft Umit Sude Böhler: Writing – original draft, validation